Principal Component Regression (PCR)

TTK 4260: Multivariate Data Analysis and ML

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Overview

- Motivation: multicollinearity and why OLS becomes unstable
- PCR algorithm: PCA on $\mathbf{X}$, regression on scores, back-projection
- Interpretation: geometry, shrinkage view, and when PCR can fail
- Model selection: choosing r and diagnostic plots (with proper CV)
- Worked example, summary, and bridge to PLS

Motivation and context

Learning Objectives

- **Synthesize** why MLR becomes unstable under multicollinearity and how PCA creates an orthogonal score space that fixes the numerical issue.
- **Carry out** the PCR algorithm: PCA on $\mathbf{X}$, regression on scores, and back-projection to coefficients in the original variable space.
- **Visualize** PCR as *projection* followed by *regression* in a reduced-dimensional space.
- **Compare** the regularization behavior of PCR (hard truncation) with Ridge (continuous shrinkage), and understand the bias–variance tradeoff.
- **Select** the number of components using cross-validation while avoiding data leakage.

Recap: Two Problems, Two Tools

Problem (MLR lecture)

- Correlated predictors $\Rightarrow$ unstable $\boldsymbol{\theta}$
- Coefficients flip signs
- Small data changes cause wild swings
- Remember the gym selfies?

Root cause:
$\mathbf{X}^\top\mathbf{X}$ is nearly singular $\Rightarrow$ $(\mathbf{X}^\top\mathbf{X})^{-1}$ explodes

Tool (PCA lecture)

- PCA finds orthogonal directions in $\mathbf{X}$
- Scores $\mathbf{T}$ are uncorrelated by construction
- Captures most variance in a few components
- Dimensionality reduction

Key property:
$\mathbf{T}^\top\mathbf{T} = \boldsymbol{\Lambda}$ (diagonal) $\Rightarrow$ no collinearity!

The Idea

What if we **regress $\boldsymbol{y}$ on the PCA scores** instead of the original correlated $\mathbf{X}$?

When Do We Need PCR?

Use PCR when. . .

- Latent-variable interpretation
- High multicollinearity
- Stable predictions
- n close to or $> m$
- Dimensionality reduction

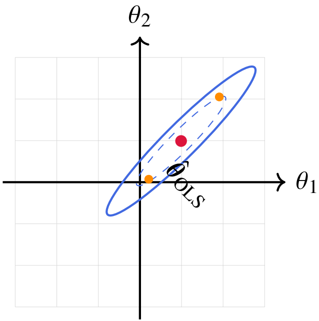
Typical applications

Spectroscopy (NIR, NMR), process monitoring, genomics, chemometrics — anywhere you have many correlated sensors / measurements and one response to predict.

The Collinearity Problem — Visually

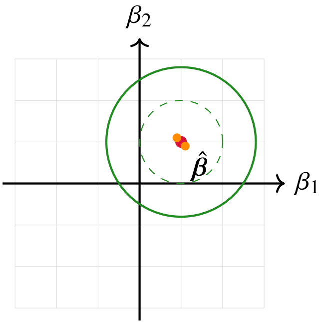
Contours of the least-squares loss $\|\boldsymbol{y} - \hat{\boldsymbol{y}}\|^2$. The ● is the optimum; ● show shifts under data perturbation[cite: 20].

OLS: Original Space (x_1, x_2)



Unstable: Elongated "valleys" mean many $\boldsymbol{\theta}$ give similar loss. Small noise slides the optimum far along the ridge[cite: 30].

PCR: Score Space (t_1, t_2)



Stable: Orthogonal scores create a "circular bowl". The optimum is uniquely defined and robust to noise.

The PCR method

PCR: Step by Step

Step 1 — Preprocess

- Center $\mathbf{X}$ and scale
- Center $\boldsymbol{y}$

Step 2 — PCA on X

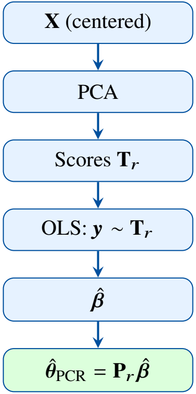
- Decompose $\mathbf{X} = \mathbf{T}\mathbf{P}^\top + \mathbf{E}$
- Keep the first r components: $\mathbf{T}_r, \mathbf{P}_r$

Step 3 — Regress $\boldsymbol{y}$ on $\mathbf{T}_r$

- OLS in score space: $\hat{\boldsymbol{\beta}} = (\mathbf{T}_r^\top \mathbf{T}_r)^{-1}\mathbf{T}_r^\top \boldsymbol{y}$

Step 4 — Back-project to original variables

- $\hat{\theta}_{\text{PCR}} = \mathbf{P}_k\, \hat{\boldsymbol{\beta}}$
- Prediction: $\hat{\boldsymbol{y}} = \mathbf{X}\,\hat{\theta}_{\text{PCR}}$



PCR in Matrix Form

Full pipeline

$$\hat{\boldsymbol{y}} = \mathbf{T}_r\,\hat{\boldsymbol{\beta}} = \underbrace{\mathbf{X}\,\mathbf{P}_r}_{\text{project to scores}}\ \underbrace{(\mathbf{T}_r^\top\mathbf{T}_r)^{-1}\mathbf{T}_r^\top\boldsymbol{y}}_{\hat{\boldsymbol{\beta}}} = \mathbf{X}\,\hat{\boldsymbol{\theta}}_{\mathrm{PCR}}$$

Expanding the pieces:

$$\hat{\boldsymbol{\beta}} = (\mathbf{T}_r^\top\mathbf{T}_r)^{-1}\mathbf{T}_r^\top\boldsymbol{y} \qquad \text{(OLS in score space — trivial, because } \mathbf{T}_r \text{ orthogonal)}$$

$$\hat{\boldsymbol{\theta}}_{\mathrm{PCR}} = \mathbf{P}_r\,\hat{\boldsymbol{\beta}} = \mathbf{P}_r\,(\mathbf{T}_r^\top\mathbf{T}_r)^{-1}\mathbf{T}_r^\top\boldsymbol{y} \qquad \text{(coefficients in original variable space)}$$

Compare with OLS:

$$\hat{\boldsymbol{\theta}}_{\mathrm{OLS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{y} \qquad \longleftrightarrow \qquad \hat{\boldsymbol{\theta}}_{\mathrm{PCR}} = \mathbf{P}_r\,(\mathbf{T}_r^\top\mathbf{T}_r)^{-1}\mathbf{T}_r^\top\boldsymbol{y}$$

Key difference

OLS inverts $\mathbf{X}^\top\mathbf{X}$ (ill-conditioned when collinear). PCR inverts $\mathbf{T}_r^\top\mathbf{T}_r$ (diagonal, well-conditioned) and restricts the solution to the subspace spanned by the first r loadings.

Why Each β_j Is a Simple Scalar

Because PCA scores are orthogonal, $\mathbf{T}_r^\top \mathbf{T}_r = \text{diag}(\lambda_1, \ldots, \lambda_r)$.
Each score-space coefficient is computed **independently**:

$$\boxed{\beta_j = \frac{\boldsymbol{t}_j^\top \boldsymbol{y}}{\boldsymbol{t}_j^\top \boldsymbol{t}_j} = \frac{\boldsymbol{t}_j^\top \boldsymbol{y}}{\lambda_j}} \qquad j = 1, \ldots, r$$

Interpretation of β_j:

- Measures **how much PC j correlates with the response $\boldsymbol{y}$**
- Large $|\beta_j|$: this PC direction is predictive
- Small $|\beta_j|$: this PC captures $\mathbf{X}$-variance but is unrelated to $\boldsymbol{y}$

No collinearity issues!

Unlike OLS, where adding/removing one variable can change *all* coefficients, each β_j in PCR is **independent** of the others. Adding PC $r+1$ does *not* change $\beta_1, \ldots, \beta_r$.

Back-Projection: From Scores to Original Variables

Once we have $\hat{\boldsymbol{\beta}} = [\beta_1, \ldots, \beta_r]^\top$, we map back:

$$\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r\, \hat{\boldsymbol{\beta}} = \sum_{j=1}^{r} \beta_j\, \boldsymbol{p}_j$$

Reading the result:

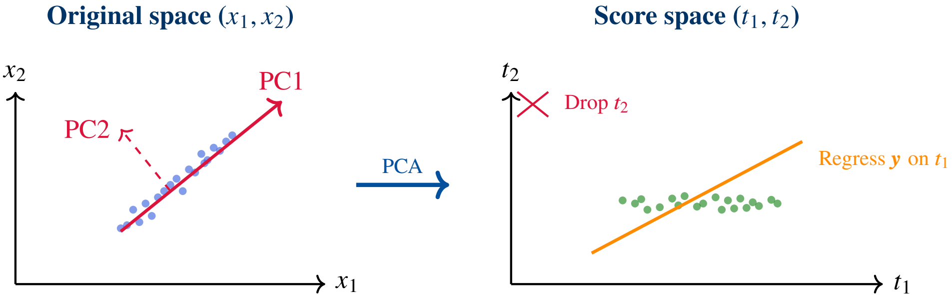
- Variable i loads heavily on a predictive PC $\Rightarrow$ large $|\hat{\theta}_i|$
- Variable i only loads on PCs unrelated to $\boldsymbol{y}$ $\Rightarrow$ small $|\hat{\theta}_i|$
- Variables with similar loading profiles get similar coefficients (grouped effect)

Practical advice

Report both $\hat{\boldsymbol{\beta}}$ (which PCs matter) and $\hat{\boldsymbol{\theta}}_{\text{PCR}}$ (which original variables matter). Together they tell the complete story.

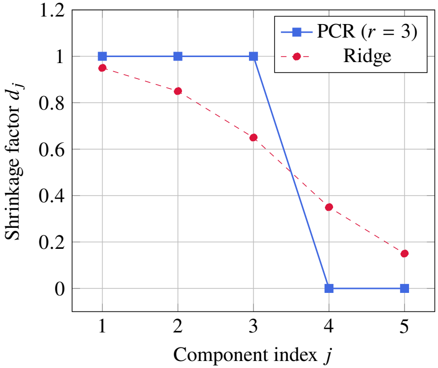
Interpretation and limitations

PCR Geometry: Project, Then Regress



PCR = rotate to PC axes $\longrightarrow$ keep high-variance directions $\longrightarrow$ regress $\boldsymbol{y}$ on those

PCR as Hard-Threshold Shrinkage



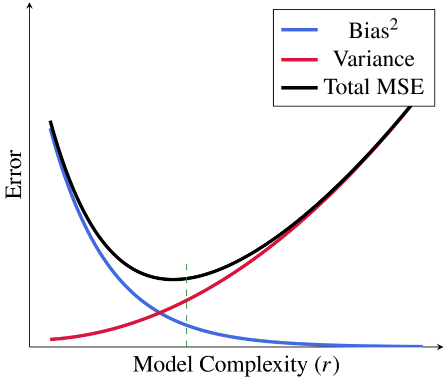
Parameters:

- j**:** The index of the Principal Component.
- d_j**:** Multiplier for the OLS solution.
- **PCR:** A binary "keep or discard" filter ($d_j \in \{0, 1\}$).
- **Ridge:** Continuous shrinkage $d_j = \frac{\lambda_j}{\lambda_j + \alpha}$.

Aggression

PCR completely removes low-variance PCs, whereas Ridge shrinks them.

Bias–Variance Tradeoff



- **Few PCs** (r small): high bias (underfitting), low variance
- **Many PCs** ($r = p$): zero bias (recovers OLS), but variance explodes if collinear
- **Cross-validation** finds the sweet spot

Head-to-Head Comparison

	OLS	**PCR**	**Ridge**
Shrinkage type	None	Hard (0 or 1)	Smooth
Bias	Zero	Moderate	Moderate
Variance	High if collinear	Low	Low
Tuning parameter	—	r (# PCs)	α (penalty)
Variable selection	No	Implicit (via PCs)	No
Interpretability	Direct $\boldsymbol{\theta}$	Via loadings $\mathbf{P}$	Direct $\boldsymbol{\theta}$
Handles $n > m$	No	Yes	Yes

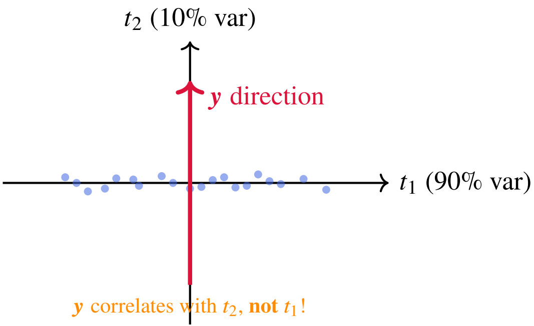
PCR wins when:

- Response lives in the top PCs
- You want latent-variable interpretation
- Strong collinearity structure

Ridge wins when:

- Predictive info is spread across many PCs
- Small PCs also carry signal
- You prefer smooth regularization

A Cautionary Example: When PCR Fails



Scenario:

- PC1 captures 90% of $\mathbf{X}$-variance
- PC2 captures only 10%
- But $\boldsymbol{y}$ is aligned with PC2!

If we set $r=1$:

- We **discard** the only predictive direction
- PCR gives poor predictions

Fundamental limitation

PCA selects directions that explain variance in $\mathbf{X}$, **not** covariance with $\boldsymbol{y}$. High $\mathbf{X}$-variance $\neq$ high predictive power.

Model selection and diagnostics

Practical Pitfall: Avoid Data Leakage in CV

Rule

When doing cross-validation, **everything** must be learned on the training fold only: centering/scaling, PCA loadings $\mathbf{P}_k$, and regression coefficients.

Wrong (leaky)

- PCA on full $\mathbf{X}$
- Then CV only the regression step
- $\Rightarrow$ optimistic RMSEP

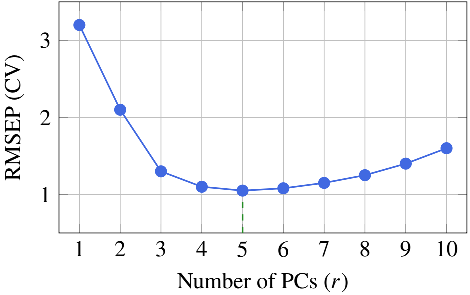
Correct (nested pipeline) For each CV split:

- Fit scaler + PCA on $\mathbf{X}_{\text{train}}$
- Transform $\mathbf{X}_{\text{train}} \longrightarrow \mathbf{T}_{\text{train}}$
- Fit regression on $\mathbf{T}_{\text{train}}$
- Transform $\mathbf{X}_{\text{val}} \longrightarrow \mathbf{T}_{\text{val}}$ using **train** PCA
- Evaluate on validation fold

Message

PCR is a **single pipeline model**. Cross-validate the pipeline, not isolated pieces.

Selecting r: Cross-Validation



Procedure:

- For $r = 1, \ldots, p$: fit PCR, evaluate RMSEP via K-fold CV
- Pick $r^* =$ minimum RMSEP

Warning

Do **not** select r based only on explained variance in $\mathbf{X}$. The most variable PCs are not necessarily the most predictive!

Explained Variance in X vs. Predictive Power

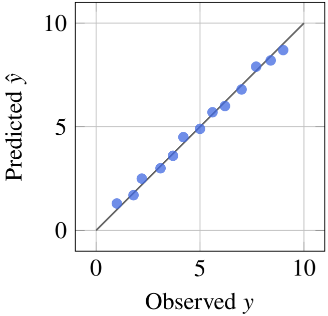
PC	X-variance (%)	Cumulative (%)	$\lvert\beta_j\rvert$ (relevance to y)
1	55	55	0.12 (weak)
2	25	80	0.85 (strong)
3	10	90	0.73 (strong)
4	5	95	0.05 (weak)
5	3	98	0.62 (moderate)

- PC1 explains the most **X**-variance but is **nearly irrelevant** for y
- PCs 2, 3, 5 are the predictive ones
- Selecting $r=1$ from a scree plot would miss the signal entirely!

Lesson

Always validate component selection against the **response** (via CV), not just the **predictors**.

PCR Diagnostic: Predicted vs Observed (CV/Test)



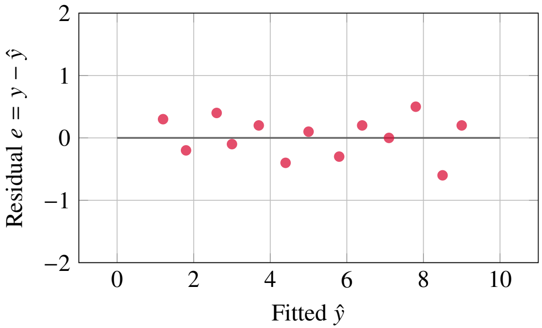
What to look for:

- Points near the 45° line $\Rightarrow$ accurate predictions
- Systematic deviation (curve) $\Rightarrow$ missing nonlinearity
- Slope < 1 often indicates shrinkage / underfitting

Reporting

Show this for **CV or a held-out test set**, not just training.

PCR Diagnostic: Residuals vs Fitted



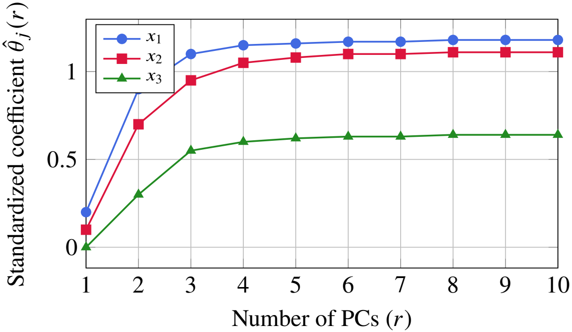
What to look for:

- Random scatter around 0 $\Rightarrow$ OK
- Funnel shape $\Rightarrow$ heteroscedasticity
- Curve / trend $\Rightarrow$ missing structure (e.g. nonlinearities)

PCR-specific note

If structure remains, adding PCs may help *only if* it captures that structure. Otherwise consider transformations or nonlinear models.

PCR Interpretation: Coefficient Paths vs Number of PCs



Why this matters:

- Shows **stability** of coefficients as model complexity grows
- If coefficients stabilize early, you can justify a smaller r
- Wild swings at large r often indicate variance inflation / noise fitting

Good practice

Plot paths for all variables, but highlight a few key ones in the slide.

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Worked example and wrap-up

Example: Process Sensors → Product Quality

Setup: 5 correlated temperature sensors ($x_1, \ldots, x_5$), predict product strength y.

OLS coefficients ($\hat{\theta}_{\text{OLS}}$):

Variable	$\hat{\theta}_j$
x_1 (Inlet)	+42.3
x_2 (Mid)	−87.6
x_3 (Outlet)	+51.2
x_4 (Jacket)	+29.8
x_5 (Ambient)	−34.1

Large magnitudes, sign flips, hard to trust.

PCR coefficients ($\hat{\theta}_{\text{PCR}}$, $k=2$):

Variable	$\hat{\theta}_j$
x_1 (Inlet)	+2.1
x_2 (Mid)	+1.8
x_3 (Outlet)	+2.3
x_4 (Jacket)	+1.5
x_5 (Ambient)	+0.4

Smooth, stable, physically sensible.

Interpretation via loadings

PC1 ≈ "overall temperature level" (all sensors load positively) ⇒ higher temperatures → stronger product. PC2 ≈ "inlet–outlet gradient" ⇒ secondary pattern.

Reading the Full PCR Story

Loadings $\mathbf{P}_r$
Which variables form each PC?

$\xrightarrow{\text{combine}}$

Model params $\hat{\boldsymbol{\beta}}$
Which PCs predict $\boldsymbol{y}$?

$\xrightarrow{\text{back-project}}$

$\hat{\boldsymbol{\theta}}_{\mathbf{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}$
Which variables predict $\boldsymbol{y}$?

"All temps load similarly on PC1"

"PC1 is predictive ($\beta_1 = 2.0$)"

"All temps get similar $\hat{\theta} \approx 2$"

Three levels of interpretation:

- **Which PCs matter?** Look at $|\hat{\beta}_j|$ — the latent patterns that predict $\boldsymbol{y}$
- **What do those PCs mean?** Look at $\mathbf{P}_r$ — loading plots from the PCA lecture
- **Which original variables matter?** Look at $\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}$ — the answer in original variable space

Summary

1. What is PCR?

- PCA on $\mathbf{X}$ to get orthogonal scores $\mathbf{T}_r$, then OLS of $\boldsymbol{y}$ on $\mathbf{T}_r$
- Back-project: $\hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\beta}}$

2. Why does it work?

- Eliminates collinearity by working in orthogonal score space
- Implicit regularization: drops low-variance (noisy) directions

3. How to choose r?

- Cross-validation on RMSEP — **not** explained variance in $\mathbf{X}$ alone

4. Compared to alternatives:

- PCR = hard-threshold shrinkage (keep/discard PCs); Ridge = smooth shrinkage
- PCR preferred when latent structure aligns with prediction; Ridge when signal is diffuse

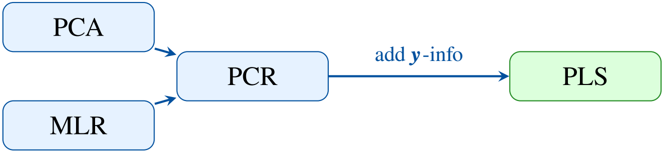
The Limitation — and What Comes Next

PCR limitation:

- PCA is **unsupervised** — it maximizes variance in $\mathbf{X}$ without looking at $\boldsymbol{y}$
- The best $\mathbf{X}$-directions may not be the best for predicting $\boldsymbol{y}$
- Can require many PCs to capture the relevant signal

Next lecture: PLS

- Finds directions that maximize **covariance** between $\mathbf{X}$ and $\boldsymbol{y}$
- **Supervised** dimensionality reduction
- Often needs fewer components than PCR
- The standard tool in chemometrics



Thank you!

Questions?

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